MICHAEL P. RAGSDALE

Accessibility Engineer | Section 508 Compliance Specialist Manual Testing • WCAG 2.1 AA Remediation • Web Platform Architecture Norfolk, VA | U.S. Citizen (Clearance Eligible / Prev. Public Trust) | hireme@michaelpragsdale.com

Professional Summary

Accessibility Engineer and Technical Specialist with deep expertise in WCAG 2.1 AA conformance, Section 508 compliance, and rigorous manual testing methodologies. Experienced in auditing enterprise web platforms, remediating complex UI components, and generating accessible documentation from legacy systems. Combines a strong background in full-stack software development with a passion for ensuring digital equity. Highly skilled in identifying contextual accessibility failures that automated scanners miss, with a track record of bridging the gap between technical requirements and inclusive user experiences.

Technical Skills

- **Accessibility Testing:** WCAG 2.1 AA, Section 508, DHS Trusted Tester methodologies, Manual Keyboard Testing, Semantic HTML Auditing
- **Assistive Tech & Tools:** NVDA, JAWS, Deque axe, WAVE, Colour Contrast Analyser, Browser Developer Tools
- **Development & Architecture:** HTML5, CSS3, Vanilla JavaScript (ES6+), PHP 8.x, JSON Routing, REST APIs
- **Infrastructure & Tracking:** Jira, Git, CI/CD, DigitalOcean, Linux, Bash

Relevant Technical Experience

Computer Operations (Section 508 Compliance Specialist)

Old Dominion University (ODU Online) | Norfolk, VA | Nov 2019 – Jul 2022

- **Enterprise Accessibility Strategy:** Served as the primary technical gatekeeper for web content and data during a massive, university-wide digital infrastructure migration to the Canvas LMS.
- **Compliance Auditing & Remediation:** Executed rigorous WCAG 2.1 AA and Section 508 compliance audits across institutional platforms. Remediated complex codebases (HTML/CSS) to ensure strict structural integrity and semantic correctness.
- **Manual vs. Automated Testing:** Utilized automated tools like Deque axe for baseline scanning while prioritizing rigorous manual testing to catch contextual failures (e.g., focus trapping, illogical reading orders, missing ARIA live regions) that scanners consistently missed.
- **Architectural Standardization:** Engineered universal, inherently accessible data and structural templates deployed across the university ecosystem, heavily streamlining workflows for instructional design faculty while enforcing baseline compliance.

Clerk-Cashier (Accessible Document Engineering)

City of Virginia Beach Parks & Recreation | Virginia Beach, VA | May 2023 – Present

- **Accessible Output Generation:** Architect and manage manual ETL workflows to extract complex operational and financial data from legacy enterprise platforms (ActiveNet) and generate strictly Section 508 compliant, screen-reader friendly digital documents.
- **Compliance Enforcement:** Govern output data to guarantee strict accessibility compliance across all generated operational documents, ensuring institutional adherence to federal and municipal accessibility mandates.
- **Training & Governance:** Serve as the primary technical trainer for municipal staff, translating complex data integrity protocols into actionable processes.

Principal Systems Architect & Founder

RaggieSoft Media | Norfolk, VA | raggiesoftmedia.com | Est. 2008 – Present Founded and operate RaggieSoft Media, an independent software development and digital media publishing ecosystem supporting web applications, cloud infrastructure, narrative publishing, music distribution, and open-source platform development.

- **Accessibility-First Platform Development:** Architected and open-sourced **Stardust Engine CMS**, an MIT-licensed, zero-database PHP MVC platform utilizing a custom Vanilla JavaScript SPA routing engine built with semantic HTML, keyboard focus trapping, and ARIA state management at its core.
- **Automated & Manual QA:** Implemented rigorous manual testing protocols (NVDA/JAWS) and automated code-review pipelines for all deployed applications, bridging the gap between full-stack architecture and inclusive design.
- **Multi-Domain & Cloud Edge:** Orchestrate a production fleet of 9 distinct domains via **Cloudflare DNS/CDN** edge routing, 301 redirection rules, and automated deployment pipelines syncing assets to **DigitalOcean Spaces** (S3-compatible object storage) and Nginx reverse proxies.
- **Automated Media Engineering:** Engineered automated server-side pipelines utilizing **FFmpeg** and Bash to transcode and normalize audio assets across multiple distribution formats, paired with custom **PHP GD** engines that dynamically composite Open Graph social cards across routes.
- **E-Commerce & Digital Fulfillment:** Integrated **Fourthwall** as an international Merchant of Record (MoR) with custom DNS architecture, powering automated fulfillment of physical merchandise and digital assets without third-party vendor lock-in.
- **AI Governance & Engineering:** Authored comprehensive architectural constraints, repository standards, and prompt protocols to govern AI-assisted engineering workflows across the codebase, ensuring maintainability, code integrity, and architectural consistency.

Education & Certification

- **Tidewater Community College | Virginia Beach, VA** Associate of Science, Information Technology (Focus: Software Engineering & Web Dev) | Expected Dec 2027
- **Old Dominion University | Norfolk, VA** Bachelor of Science, Customized Studies in Leadership | Prior Coursework
- **Wilson Workforce and Rehabilitation Center | Fishersville, VA** Certificate, Business & Information Technology | 2011